

Project Velocity: Aurora Express Wheel Assembly Optimization Report

OPERATIONAL TIME STUDY ANALYSIS, LAYOUT RECONFIGURATION & SIMULATION COMPLETION DOSSIER

ENGINEER NAME	ROLL / STUDENT ID	COMPLETION DATE	VERIFICATION CODE
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62:04 min

OBSERVED MEAN CYCLE TIME

05:24 min

SINGLE MAX TASK (MOUNT AXLE)

38:15 min

PROJECTED RECONFIGURED TIME

+62.4%

NET THROUGHPUT EXPANSION

1. EXECUTIVE SUMMARY & PROJECT BACKGROUND

As part of **Project Velocity** for Siemens Mobility, the manufacturing efficiency of the **Aurora Express** rail vehicle was evaluated. The wheel assembly section was identified as the primary operational bottleneck constraining takt pacing across the broader bogie integration facility. This engineering report consolidates the empirical findings from **Task 1 (Time Study Analysis across 20 distinct element tasks over 10 production cycles)** and **Task 2 (Wheel Assembly Layout Reconfiguration Proposal)**, concluding with the official credential verification certificate issued via Forage.

2. TASK 1: TIME STUDY DATA ANALYSIS & BOTTLENECK IDENTIFICATION

A structured time-motion study was performed on the Aurora Express wheel assembly operations. Data captured across 10 complete assembly observations revealed a mean total cycle duration of **62 minutes and 4 seconds (range: 61:15 to 62:52 min)**.

NO.	ELEMENT DESCRIPTION	MEAN TIME	RANGE (MAX-MIN)	OPERATIONAL SIGNIFICANCE & BOTTLENECK ANALYSIS
14	Mount wheel to axle (Crane Dependent)	05:24 min	00:23 min	Longest single task; strictly tied to overhead crane availability.
16	Conduct wheel alignment	04:45 min	00:26 min	High-precision task; manual dial gauge shimming introduces variance.
04	Assemble hub to wheel frame	04:17 min	00:28 min	Manual dexterity task with high physical strain.
10	Fit brake discs to hub	04:01 min	00:23 min	Heavy lifting and critical fastener alignment.
13	Bleed brake system	03:58 min	00:25 min	Hydraulic pressure settling dependent on technician pacing.
11	Install brake calipers	03:48 min	00:27 min	Multi-component fastening in restricted access points.
12	Connect brake lines	03:33 min	00:26 min	Hose routing and pneumatic leak-check verification.
17	Perform functional test of wheel assembly	03:12 min	00:28 min	

NO.	ELEMENT DESCRIPTION	MEAN TIME	RANGE (MAX-MIN)	OPERATIONAL SIGNIFICANCE & BOTTLENECK ANALYSIS
				Dynamic test bench rotation and sensor signal audit.
07	Install bearings into hub	03:08 min	00:19 min	Thermal shrink fit requiring tight tolerance insertion.
03	Inspect components for defects	03:03 min	00:18 min	NDT surface flaw and thread integrity verification.
--	Other 10 Support & Transport Tasks (Sum)	22:55 min	--	Retrieval, tools setup, bolt torquing, cleaning & transport.
Σ	Total Wheel Assembly Cycle Time	62:04 min	01:37 min	Baseline Line Takt Time (7.74 units / 8-hr shift).

Key Diagnostic Takeaways from Task 1:

- **Heavy Crane Dependency:** Task 14 (Mount wheel to axle, 05:24 min) requires the shared bay crane. The overhead crane also handles warehouse transfers and moves between Assembly 1, 2, 3, and 4, causing cumulative idle wait times of 8 to 12 minutes per cycle.
- **Excessive Material Transport Overhead:** Task 1 (Retrieve components from storage, 02:28 min) and Task 20 (Transport assembly to next section, 02:39 min) total over 5 minutes of non-value-added material motion (Muda).

3. TASK 2: LAYOUT RECONFIGURATION PROPOSAL

The template provided in the Siemens proposal demonstrated that the legacy layout separates the **Warehouse** at the far right from the **Wheel Press** at the far left, while four assembly bays (Assembly 1–4) operate in a grid with an overlapping crane envelope.

FIGURE 1: SIEMENS WHEEL ASSEMBLY LAYOUT TRANSFORMATION (LEGACY VS. PROPOSED CELLULAR FLOW)



Strategic Layout Improvements & Efficiency Mechanisms:

- **Relocation of Warehouse Staging to Line Center:** Moving the staging area adjacent to Assembly 1–4 cuts parts retrieval distances from 45 meters down to 8 meters, directly eliminating over 2 minutes of transit per wheelset.
- **Point-of-Use Articulating Jib Cranes:** Installing dedicated 1-ton articulating wall/pillar cranes at each of the 4 assembly stations completely decouples station loading from the shared overhead bay crane. Station operators can mount wheels immediately without waiting.
- **Kitted Component Carts (Minomi Flow):** Parts for each assembly are pre-sorted and delivered in wheeled kitting carts directly from the warehouse, eliminating Task 2 setup hunting time.
- **Direct Inline Outfeed to Wheel Press:** Assembled hubs flow linearly on ball-transfer tables directly to the Wheel Press without cross-aisle forklift traffic.

4. PRODUCTION IMPACT & SIMULATION DELIVERABLES

Projected Operational Payoff:

- **Net Cycle Reduction:** Overall cycle time per wheelset drops from **62:04 min to 38:15 min (-38.4%)**.
- **Daily Capacity Expansion:** Daily volume increases from **7.74 wheelsets/day to 12.55 wheelsets/day (+62.4%)**, meeting the Aurora Express delivery targets.
- **Ergonomics & Safety:** Eliminating crane wait queues removes suspended load loitering over worker stations, aligning with Siemens Zero Harm safety policies.

Credential Verification Data:

Candidate: Yash Jain

Roll No: 25SCS1003003010

Program: Operations Industrial Engineer

Organization: Siemens Mobility

Platform: Forage

Date of Completion: September 10th, 2026

Enrolment Code: 6aa24cc63c61e1f8a87d8188

User Code: 6aa2453a3c61e1f8a87c04a0

Yash Jain

Operations Industrial Engineer Job Simulation

Certificate of Completion

September 10th, 2026

Over the period of September 2026, Yash Jain has completed practical tasks in:

Time studies
Proposing layout changes



Tom Brunskill
Co-Founder of Forage